

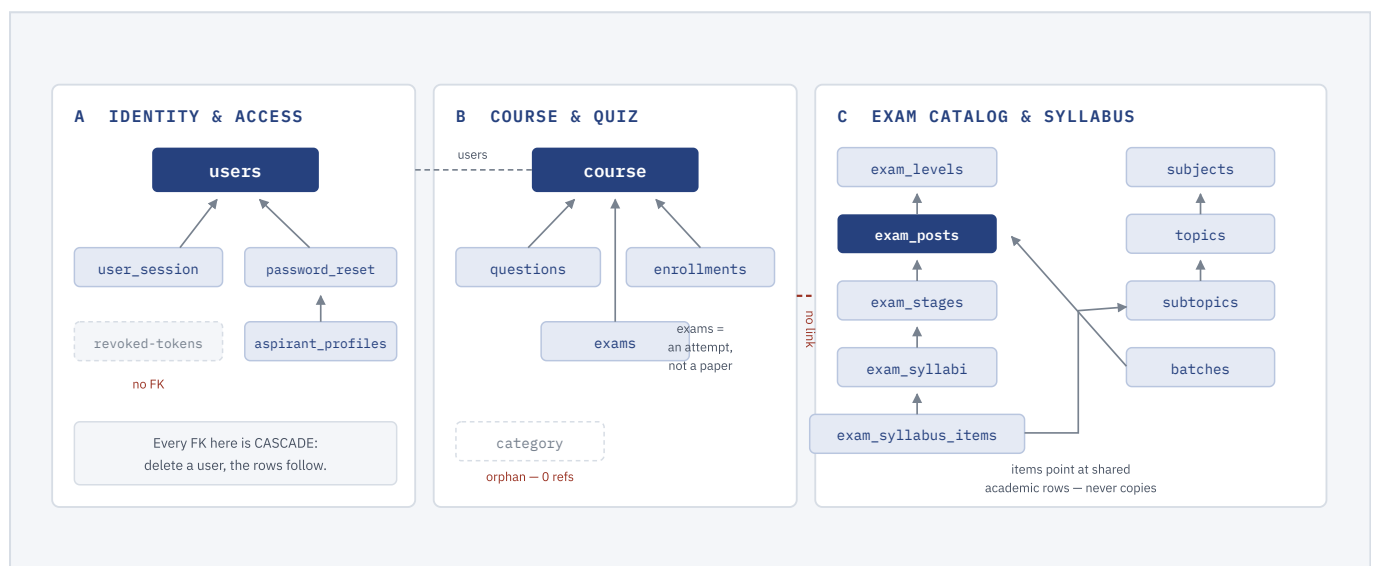
How the PSC Tables Relate

Nineteen tables in three loosely-joined groups. This document traces every foreign key, what each one does on delete, and the four places where the schema does not constrain what the code assumes.

Source `src/**/*.entity.ts` Tables 19 Foreign keys 17 Generated 9 Sep 2026

01 The three groups

The schema is not one connected graph. It is three clusters that meet only at **users** and **course**. Knowing which cluster a table belongs to explains most of its delete behaviour and most of its naming.



Group **A** hangs entirely off **users**. Group **B** is the older course-and-quiz track, hanging off **course** and **users**. Group **C** is the newer PSC exam catalog, joined to a shared academic hierarchy by the syllabus. **B** and **C** never touch — which is the single most consequential fact about this schema.

02 Group A • Identity and access

Four tables orbit **users**. Three carry a real foreign key with **CASCADE**; one does not carry a foreign key at all.

CHILD TABLE	COLUMN	PARENT	CARDINALITY	ON DELETE	MEANING
user_session	user	users	many → 1	NO ACTION	One row per login. Holds the refresh token, device string and a revoked flag.
password_reset_tokens	userId	users	many → 1	CASCADE	Single-use reset token with an expiry and an isUsed flag.
aspirant_profiles	user_id	users	1 → 1	CASCADE	PSC-specific detail: date of birth, community category, Malayalam proficiency. user_id is unique, so a user has at most one.
revoked-tokens	userId	—	—	no constraint	Logout blacklist. userId is a plain integer column with no foreign key, so nothing stops an orphan.

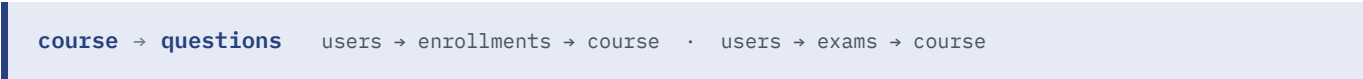
SOFT DELETE IS NOT WHAT TYPEORM THINKS IT IS

users.deletedAt is a plain `@Column`, not a `@DeleteDateColumn`. TypeORM therefore does **not** filter it automatically — every query has to pass `deletedAt: IsNull()` by hand, and any query that forgets will return deleted users.

This already leaks: `GET /aspirant-profiles` joins `relations: ['user']`, and a soft-deleted user still arrives attached to their profile. Every other soft-deleted table in this schema uses a real `@DeleteDateColumn` and filters itself.

03 Group B • Course and quiz

The original track. A course holds questions; users enrol in courses and sit attempts against them.



CHILD TABLE	COLUMN	PARENT	ON DELETE	MEANING
questions	courseId	course	CASCADE	Deleting a course destroys its questions.
questions	createdBy	users	SET NULL	Authorship survives the author. The only SET NULL in the schema.
questions	updatedBy	users	SET NULL	Same, for the last editor.
enrollments	userId	users	CASCADE	Join table between a user and a course, with a status of pending / active / completed / cancelled. Note there is no unique constraint on the pair, so a user can be enrolled in the same course twice.
enrollments	courseId	course	CASCADE	
exams	userId	users	CASCADE	One attempt session. Carries questionIds as a JSON array, the submitted answers map, and the score.
exams	courseId	course	CASCADE	
category	—	—	n/a	No foreign keys in either direction. Nothing in the codebase references it; a seed script migrates its contents into subjects.

TWO DIFFERENT THINGS ARE BOTH CALLED AN EXAM

exams in this group is an **attempt session** — one user's run through a set of questions. The catalog entity in Group C is **exam_posts**, named that way precisely because **exams** was already taken and holds live data.

The foreign keys pointing at **exam_posts** are still named **exam_id**, matching the domain language. So **exam_stages.exam_id** means an exam *post*, while **exams.id** means an attempt. They are unrelated.

04 Group C · Exam catalog, syllabus, academic content

Two independent hierarchies joined by a syllabus. The catalog descends from a qualification band to a single stage of a single post; the academic hierarchy is global and shared by every exam that needs it.

`exam_levels` → `exam_posts` → `exam_stages` → `exam_syllabi` → `exam_syllabus_items` ←
`subjects` / `topics` / `subtopics`

CHILD TABLE	COLUMN	PARENT	ON DELETE	MEANING
<code>exam_posts</code>	<code>exam_level_id</code>	<code>exam_levels</code>	RESTRICT	A post — Sub Inspector, LDC — sits under a qualification band such as Degree Level.
<code>exam_stages</code>	<code>exam_id</code>	<code>exam_posts</code>	RESTRICT	Prelims, Mains, Physical Test. Stages are data, so a post may have any number in any order.
<code>exam_syllabi</code>	<code>exam_stage_id</code>	<code>exam_stages</code>	RESTRICT	At most one live syllabus per stage, enforced by a partial unique index. The syllabus hangs off the stage, not the post, because prelims and mains cover different ground.
<code>exam_syllabi</code>	<code>exam_id</code>	<code>exam_posts</code>	RESTRICT	Denormalised on purpose, so a whole post's syllabus lists without a join. The service rejects a stage that does not belong to the given post, keeping the two consistent.
<code>exam_syllabus_items</code>	<code>syllabus_id</code>	<code>exam_syllabi</code>	CASCADE	Delete a syllabus and its lines go with it.
<code>exam_syllabus_items</code>	<code>subject_id</code>	<code>subjects</code>	RESTRICT	Required. Subject alone means the whole subject is in the syllabus.
<code>exam_syllabus_items</code>	<code>topic_id</code>	<code>topics</code>	RESTRICT	Nullable. Present, it narrows the item to one topic.
<code>exam_syllabus_items</code>	<code>subtopic_id</code>	<code>subtopics</code>	RESTRICT	Nullable. Present, it narrows the item to a single subtopic.
<code>topics</code>	<code>subject_id</code>	<code>subjects</code>	RESTRICT	Fundamental Rights under Indian Constitution.
<code>subtopics</code>	<code>topic_id</code>	<code>topics</code>	RESTRICT	Article 21 under Fundamental Rights. The leaf of the hierarchy.
<code>batches</code>	<code>exam_id</code>	<code>exam_posts</code>	RESTRICT	A coaching cohort targeted at one post, with a mode, dates and a head count.

Why the academic tables are not owned by an exam

“Indian Constitution” is **one** row in `subjects`. The Degree Level, Plus Two Level, SI and LDC syllabi all reach it through `exam_syllabus_items`. There is no per-exam copy such as “SI Constitution”. That is what makes the item table a bridge rather than a child.

Three unique indexes on one table

An item can be mapped at three depths, so `topic_id` and `subtopic_id` are nullable. Postgres treats NULLs in a unique index as distinct values, so a single index over those columns would happily accept duplicates. The table therefore carries three partial unique indexes, one per depth — subject only, subject + topic, and subject + topic + subtopic.

05 Delete semantics at a glance

Which rows disappear, which block the delete, and which are only marked gone.

BEHAVIOUR WHERE

CASCADE aspirant_profiles, password_reset_tokens, enrollments, exams, questions→course, exam_sy

RESTRICT the whole of Group C – exam_posts, exam_stages, exam_syllabi, topics, subtopics, batche

SET NULL questions.createdBy, questions.updatedBy

none revoked-tokens.userId, user_session.user, every created_by / updated_by / deleted_by au

Soft versus hard delete

MODE TABLES

SOFT course, aspirant_profiles, exam_levels, exam_posts, exam_stages, exam_syllabi, subjects, top

SOFT users

HARD exam_syllabus_items, enrollments, exams, questions, category, user_session, revoked-tokens,

06 Where the schema does not match the intent

Four gaps that the foreign keys cannot express, ordered by what they cost.

1. **Group B and Group C are not connected.** An attempt (exams) points at a `course`, never at an `exam_stage`. A student can sit “a random 30 from course X”, but not “the SI Preliminary paper”. The catalog describes exams nobody can yet take, and the attempt engine runs exams the catalog does not know about. Closing this is the single largest piece of work the schema implies.
2. **Negative marking is modelled but never applied.** `exam_stages.negative_mark` exists and holds real values, which is how PSC papers actually score. The grading code in the attempt engine never reads it, because it cannot reach a stage from an attempt.
3. **Question selection has no referential integrity.** `exams.questionIds` is a JSON array of integers, not a join table. A question can be deleted while an attempt still references it; nothing at the database level objects.
4. **Audit columns are unconstrained.** Every table in Group C carries `created_by`, `updated_by` and `deleted_by` as plain `int`. They hold user ids taken from the JWT but have no foreign key, so they can point at users who never existed.

Derived from the TypeORM entity definitions under `src/**/*.entity.ts`. The schema is created by `synchronize: true` – there are no migration files, so the entities are the only source of truth.